



**Ahmedabad**  
University

## **ECE 501: Digital Image Processing**

### **Week 2 Progress Report**

Group 6

Vehicle Orientation Estimation Using Conventional Image Processing

Submitted to faculty: Prof. Mehul Raval

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#### Student Details

| <b>Roll No.</b> | <b>Name of the Student</b> | <b>Email Id</b>          |
|-----------------|----------------------------|--------------------------|
| AU2440098       | Dhruvika Mehta             | dhruvika.m@ahduni.edu.in |
| AU2440103       | Krisha Desai               | krisha.d@ahduni.edu.in   |
| AU2440276       | Manthan Pancholi           | manthan.pl@ahduni.edu.in |
| AU2440262       | Rim Shah                   | rim.s@ahduni.edu.in      |

GitHub Repository (DRASHTI-HaOBB):

<https://github.com/ERYAGNIK003/DRASHTI-Dataset>

Dataset Source:

DRASHTI-HaOBB – Heading-angle Oriented Bounding Box Dataset

<https://zenodo.org/records/18278989>

## **1. Work Done This Week**

In Week 2, we moved from understanding the project and dataset to practical data processing.

We downloaded the DRASHTI-HaOBB original training annotations and the framewise information CSV. Since the training image file is very large, we used Google Drive and Google Colab for accessing and processing the required data.

We studied the annotation files and verified the information provided for each vehicle, including the four OBB corner coordinates, class, difficulty and heading angle. The image and annotation files were matched using their file names.

A Python-based annotation parser was created in Google Colab to read the annotation information. Using the OBB coordinates, we developed a preliminary method to extract individual vehicle crops from the images.

First, the process was tested on one vehicle image to verify that the correct image, annotation and OBB were matched. After this, the same process was applied to ten selected vehicle samples. The extracted crops were visually checked to make sure that the correct vehicles were obtained and that their original orientation was preserved.

The ground-truth heading values were also recorded and prepared for conversion from the 0°–359° range to the 0°–90° orientation range required for the project. A CSV file was created containing the vehicle class, difficulty, heading, orientation and crop dimensions.



*Figure 1. Example of a DRASHTI-HaOBB training image with the annotated oriented bounding box (OBB) used to identify the vehicle region*

Auto3WCargo | GT Heading = 270.0° | GT Orientation = 90.0°



*Figure 2. Example of an extracted vehicle crop from DRASHTI-HaOBB with its ground-truth heading and corresponding 0°–90° orientation*



*Figure 3. Sample vehicle crops extracted from DRASHTI-HaOBB OBB annotations, showing preservation of the original vehicle orientation*

|    | A                  | B              | C          | D        | E           | F          | G          | H              | I          | J           | K                   |
|----|--------------------|----------------|------------|----------|-------------|------------|------------|----------------|------------|-------------|---------------------|
| 1  | vehicle_uid        | image_item     | vehicle_id | class_id | class_name  | difficulty | gt_heading | gt_orientation | crop_width | crop_height | crop_path           |
| 2  | fajh_DDXl_sVyD_vef | fajh_DDXl_sVyD | 103        | 0        | Auto3WCargo | 0          | 75         | 75             | 183        | 244         | /content/week2_10_1 |
| 3  | w8fp_m2WO_Ehux     | w8fp_m2WO_Ehux | 3          | 0        | Auto3WCargo | 0          | 90         | 90             | 72         | 130         | /content/week2_10_1 |
| 4  | hW19_22e2_Ik0_vef  | hW19_22e2_Ik0  | 5          | 0        | Auto3WCargo | 0          | 173        | 7              | 166        | 79          | /content/week2_10_1 |
| 5  | pYWE_TnGD_nEIC     | pYWE_TnGD_nEIC | 36         | 0        | Auto3WCargo | 0          | 80         | 80             | 132        | 175         | /content/week2_10_1 |
| 6  | lv6a_dp9T_xWho_vef | lv6a_dp9T_xWho | 27         | 0        | Auto3WCargo | 0          | 94         | 86             | 113        | 197         | /content/week2_10_1 |
| 7  | OaRn_LJEV_37zf_ve  | OaRn_LJEV_37zf | 53         | 0        | Auto3WCargo | 0          | 347.1      | 12.9           | 129        | 84          | /content/week2_10_1 |
| 8  | Yme_u19R_I0V5_ve   | Yme_u19R_I0V5  | 45         | 0        | Auto3WCargo | 0          | 274        | 86             | 227        | 322         | /content/week2_10_1 |
| 9  | 4b3m_Gv8b_e30r_v   | 4b3m_Gv8b_e30r | 54         | 0        | Auto3WCargo | 0          | 270        | 90             | 103        | 204         | /content/week2_10_1 |
| 10 | 7xQC_Xode_PVRI_v   | 7xQC_Xode_PVRI | 63         | 0        | Auto3WCargo | 0          | 347.1      | 12.9           | 130        | 84          | /content/week2_10_1 |
| 11 | u7ym_lo7b_z4I6_vef | u7ym_lo7b_z4I6 | 71         | 0        | Auto3WCargo | 0          | 350        | 10             | 124        | 79          | /content/week2_10_1 |

Table 1. Metadata of the ten vehicle samples extracted for the initial Week-2 analysis

The overall data-processing pipeline completed this week was as below:

Dataset → Annotation Parsing → Image/Annotation Matching → OBB-based Crop Extraction  
→ Ground-truth Information Preparation.

## 2. Literature Review

We continued the preliminary literature review on conventional image-processing methods for orientation estimation.

The main methods considered were PCA, contour-based and geometric approaches, and the Hough Transform. PCA can be used to find the dominant direction of a vehicle shape, while contours and geometric features can help describe its main axis. Hough Transform can be used to detect strong lines and edges.

These methods are being considered because the project requires conventional image-processing techniques. Further testing on actual vehicle crops is needed before selecting the final method.

## 3. Work to Be Done Next Week

In Week 3, we plan to start the actual orientation-estimation stage of the project.

The main tasks will be:

- Apply grayscale conversion, smoothing and edge detection to the vehicle crops.
- Extract contours or other shape features.
- Implement a preliminary PCA-based orientation estimation method
- Explore contour/geometric and Hough Transform based approaches

- Compare estimated orientation with the ground-truth orientation
- Define the angular-error calculation
- Test the methods on a larger number of vehicle samples and study the effect of vehicle size and image quality.

#### **4. Challenges Faced**

The main challenge this week was the large size of the training image dataset, which made direct downloading difficult. Google Drive and Google Colab were therefore used to access and process the required data.

Another challenge was correctly matching the image files with their annotation files and interpreting the four OBB coordinates.

The conversion of the  $0^{\circ}$ – $359^{\circ}$  heading angle to the required  $0^{\circ}$ – $90^{\circ}$  orientation also needs to be applied consistently.

The vehicle samples also differ in size, shape and image quality, which may affect the performance of conventional image-processing methods.

#### **5. Current Status**

The dataset and annotation-processing stage has been completed successfully.

We have:

- Parsed the DRASHTI-HaOBB annotations.
- Matched images with their corresponding annotations.
- Extracted vehicle crops using OBB coordinates.
- Tested the crop extraction on one image and then on ten vehicle samples.
- Recorded the ground-truth heading, orientation and vehicle metadata.

The orientation-estimation method has not yet been finalized. In the next stage, we will begin testing PCA, contour/geometric and Hough Transform based methods on the extracted crops.

## 6. References

- [1] Y. Bhavsar, M. Zaveri, M. Raval, K. R. Patel, and S. B. Zaveri,  
“Descriptor: Drone Nadir-view Annotated Images of Vehicles Detection Dataset for India with  
Heading-angle Oriented Bounding Box (DRASHTI-HaOBB),” *IEEE Data Descriptions*, 2026,  
doi: 10.1109/IEEEDATA.2026.3670752.
  
- [2] Y. Bhavsar, M. Zaveri, M. Raval, S. Zaveri, and Ahmedabad University,  
“DRASHTI-HaOBB: Drone nadiR-view Annotated imageS of veHicles dataseT for India -  
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